CHANDIGARH UNIVERSITY

University Institute of Computing (UIC)

MINI PROJECT REPORT ON

LEARNING MANAGEMENT SYSTEM (DevLMS)

Submitted by

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1. OBJECTIVE

The primary objective of this project is to design and develop a fully functional Frontend for a Developer Learning Management System (DevLMS) using modern web technologies. The project demonstrates proficiency in HTML5, CSS3, JavaScript, and the Bootstrap 5 framework to build a real-world, responsive, and interactive web application.

Specific Objectives

- Design a multi-page responsive SPA-style frontend with sidebar navigation and consistent layout.
- Implement user authentication flow with login and registration forms.
- Build a dynamic course listing and course detail page with topic-wise content navigation.
- Integrate a live code editor with real-time HTML/CSS/JS preview pane.
- Create an interactive quiz module with radio-based question rendering and score computation.
- Develop a user profile management page with data update and account deletion capabilities.
- Implement a RESTful API integration layer using a centralized fetch-based api.js utility.
- Apply glassmorphism UI design, CSS custom properties, dark mode support, and smooth animations.

2. INTRODUCTION

DevLMS (Developer Learning Management System) is a comprehensive, frontend-driven web platform built to help aspiring developers learn programming through structured courses, interactive coding practice, and topic-wise quizzes. Inspired by modern e-learning platforms like Udemy and Coursera, DevLMS offers a clean, professional interface that prioritizes usability, performance, and visual clarity.

The application is built entirely using frontend web technologies — HTML5, CSS3, Vanilla JavaScript, and Bootstrap 5 — and communicates with a backend REST API (Node.js/Express) for data persistence. The project follows a modular architecture where each page is self-contained yet shares a common design language, navigation sidebar, and global API utility.

Key Features of the Platform

- Home / Landing Page — Course browsing with authentication-aware CTA and filter options.
- Authentication Page — Tabbed login and registration with token-based JWT authentication.

- Dashboard — Personalized statistics showing completed topics, quiz averages, and code practice hours.
- Courses Page — Searchable grid of courses with thumbnail, category badges, and navigation links.
- Course Detail Page — Topic sidebar with dynamic content panel showing theory, videos, and quiz links.
- Code Practice (Editor) — Full-screen live HTML/CSS/JS editor with instant preview and cloud save.
- Quiz Module — Styled multiple-choice assessments with real-time option selection and score display.
- Profile Page — User info management with avatar initials, editable fields, and account deletion modal.

The project showcases real-world frontend engineering practices including CSS custom properties for theming, glassmorphism effects, responsive grid layouts, event-driven JavaScript, local storage for session management, and progressive UI feedback (loaders, toasts, spinners).

3. SOFTWARE & TECHNOLOGY USED

The following technologies, libraries, and tools were used during the design and development of the DevLMS frontend:

Table 1 SOFTWARE & TECHNOLOGY USED

Technology	Description	Usage in Project
HTML5	HyperText Markup Language (v5)	Page structure, semantic elements, forms, iframes
CSS3	Cascading Style Sheets (v3)	Custom properties, glassmorphism, animations, dark mode
JavaScript (ES6+)	Vanilla JavaScript (ECMAScript 2015+)	DOM manipulation, async/await, event handling, fetch API
Bootstrap 5.3	CSS component framework by Twitter/Meta	Grid layout, cards, modals, badges, nav pills, spinners
Font Awesome 6	Icon library (CDN)	Sidebar icons, action buttons, status indicators
Google Fonts	Web font service	Inter and Outfit fonts for professional typography
Fetch API	Native browser API for HTTP requests	Centralized api.js for GET, POST, PUT calls to backend

LocalStorage	Browser key-value storage	JWT token and user data persistence across sessions
Visual Studio Code	Source code editor by Microsoft	Primary development IDE with Live Server extension

Development Environment

- Operating System: Windows 11 / macOS / Ubuntu Linux
- Code Editor: Visual Studio Code (v1.90+) with ESLint, Prettier, Live Server extensions
- Browser: Google Chrome 124+ (primary), Firefox 126+ (cross-browser testing)
- Version Control: Git 2.x with GitHub repository hosting
- API Testing: Postman / Thunder Client VS Code Extension

4. PROJECT STRUCTURE

The DevLMS project follows a clean, modular folder structure that separates concerns between HTML pages, CSS styling, and JavaScript modules:

DevLMS-frontend/

```
├── index.html — Landing page with course listing & auth CTA
├── login.html — Login & registration tabbed interface
├── dashboard.html — User stats and progress overview
├── courses.html — Searchable course grid listing
├── course-detail.html — Topic sidebar with content & quiz navigation
├── editor.html — Full-screen live code editor with preview
├── quiz.html — Multiple-choice quiz assessment module
├── profile.html — Profile management with delete account modal
├── css/
│       └── style.css — Global stylesheet with CSS variables, themes
└── js/
        ├── api.js — Centralized Fetch API utility (GET/POST/PUT)
        ├── auth.js — Login & registration form handlers
```

├── dashboard.js — Dashboard data loading & rendering

└── editor.js — Code editor initialization (legacy ref)

5. INPUT SCREENSHOTS

The following section describes the input screens of the DevLMS application — pages where users provide data or interact with the system to access content.

5.1 Landing Page (index.html)

The landing page serves as the entry point for unauthenticated users. It features a hero section with a gradient radial background, a prominent CTA button, and a dynamically rendered course grid loaded via the API. Users can browse available courses and filter by category. If already logged in, the navbar updates to show 'Go to Dashboard' instead of the Login link.

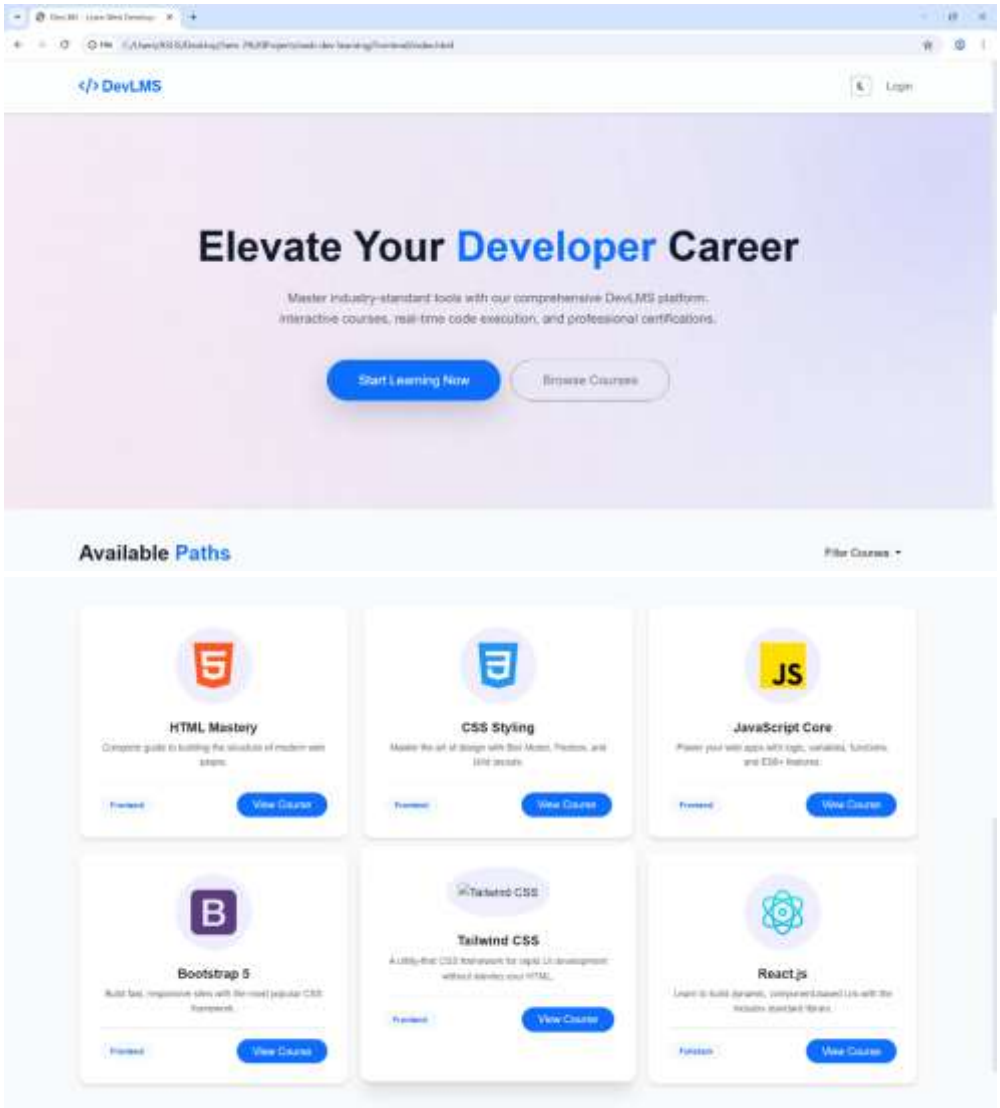

Figure 1 Landing Page

Key Input Elements: Category filter dropdown, 'Start Learning Now' CTA button, individual course 'View Course' buttons (redirect to login if unauthenticated).

5.2 Login & Registration Page (login.html)

The authentication page provides a dual-tab interface — Login and Register — using Bootstrap Nav Pills. The Login form accepts an email or mobile number along with a password. The Register form collects full name, email, mobile, password, and confirm password fields. Client-side validation ensures password matching before submission. JWT tokens are stored in localStorage upon successful authentication.

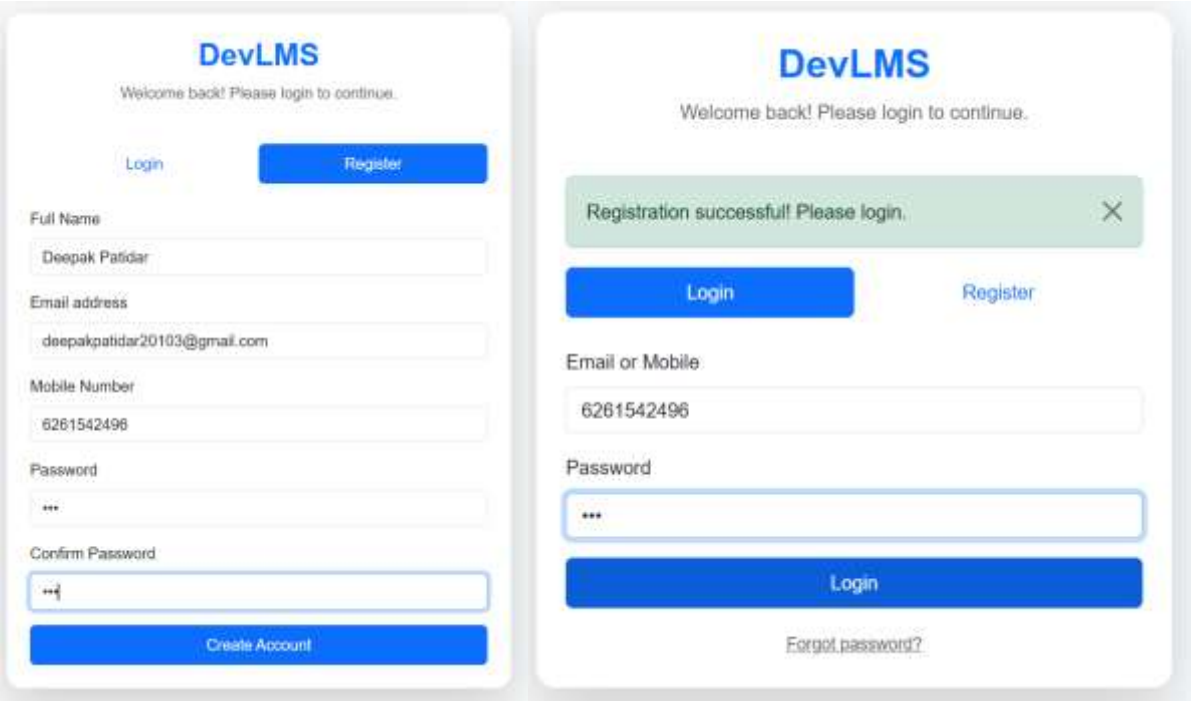

Figure 2 Login & Registration Page

Key Input Elements: Email/mobile input, password field, 'Login' submit button, 'Full Name', 'Email', 'Mobile' and 'Confirm Password' fields in the registration tab.

5.3 Course Detail Page (course-detail.html)

After selecting a course, students navigate to the detail page which dynamically loads all topics from the API based on the course ID in the URL query string. The left column displays a scrollable list of topics; clicking any topic renders its theory content and optional embedded video in the right panel. A 'Take Topic Quiz' button links to the quiz module for that specific topic.

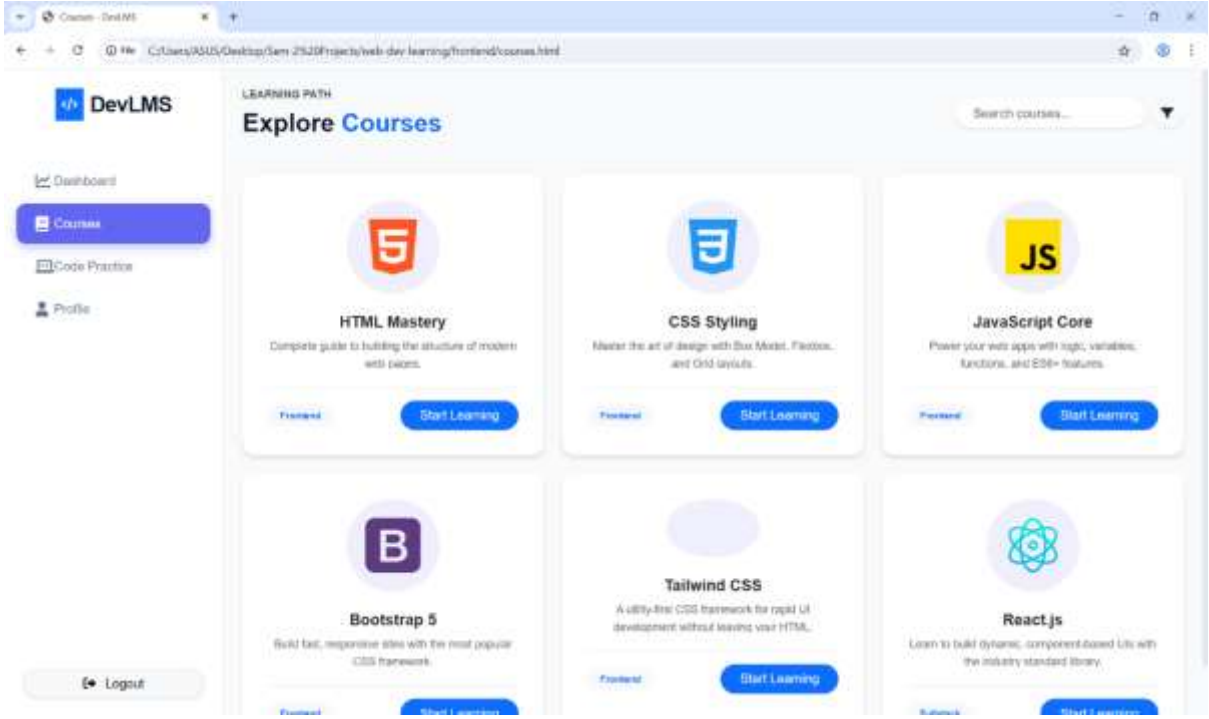

Figure 3 Course Detail Page

Key Input Elements: Topic list (clickable), 'Practice Code' link to editor, 'Take Topic Quiz' button with dynamic topic ID parameter.

5.4 Code Editor (editor.html)

The live code editor occupies the full viewport and is split into two equal panels. The left panel contains a styled dark-themed textarea where users write HTML, CSS, and JavaScript. The right panel is an iframe that renders the code in real-time when the 'Run' button is clicked. Tab key support is implemented for proper indentation. The 'Save' button posts the code snippet to the backend API.

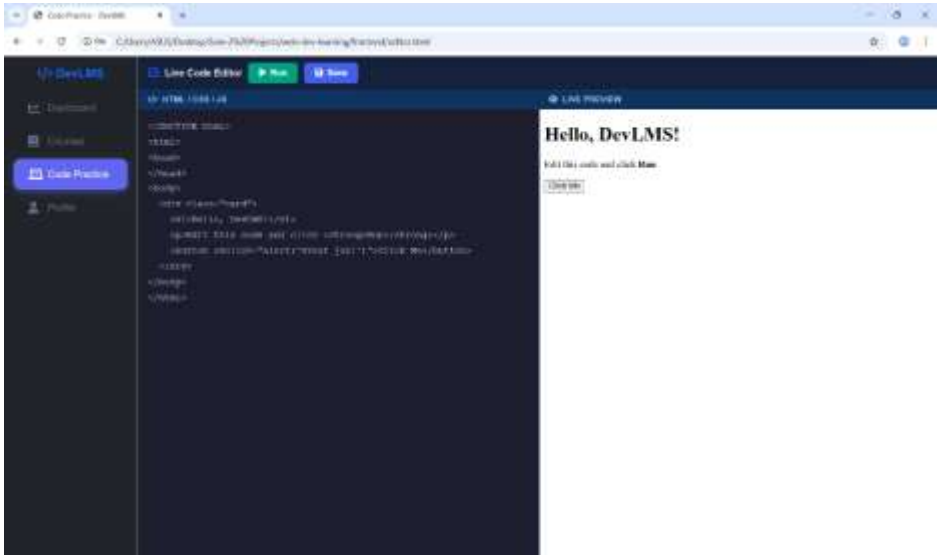

Figure 4 Code Editor

6. OUTPUT SCREENSHOTS

The output screens display the results of user actions — data processed and rendered by the system after inputs are submitted.

6.1 Dashboard (dashboard.html)

The dashboard is the main authenticated home screen. It displays three stat cards: Completed Topics (with progress bar), Quiz Performance (average percentage), and Practice Hours (number of code submissions). Below the cards, a responsive table shows recent quiz activity with topic name, course name, date, progress bar, and score badge. All data is fetched from the /api/user/profile endpoint.

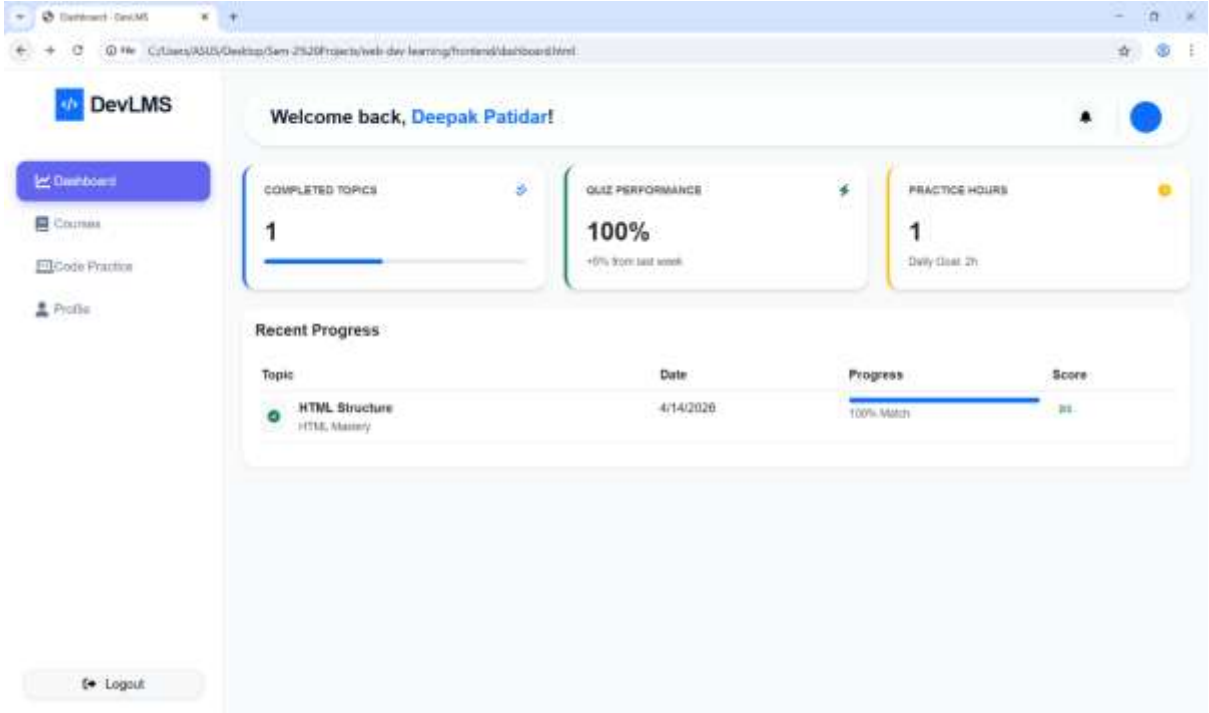

Figure 5 Dashboard

Output displayed: Topics count, quiz average %, code submission count, and a tabular history of quiz scores with visual progress bars.

6.2 Quiz Result Screen (quiz.html)

Upon completing a quiz and clicking 'Complete Assessment', the quiz form is hidden and the result container animates into view. It displays a gold trophy icon, a congratulatory message, and the score in a large, bold format (e.g., 'You scored 7 out of 10'). The result is simultaneously posted to the backend via the /api/quizzes/submit endpoint for persistence in the user's progress record.

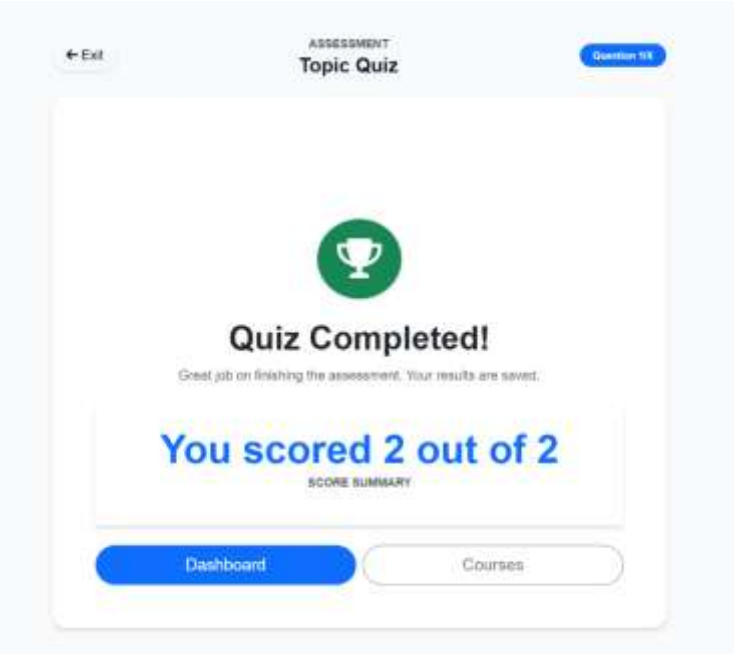

Figure 6 Quiz Result Screen

6.3 Profile Page (profile.html)

The profile page renders the authenticated user's data loaded from the API. The avatar section shows generated initials from the user's full name. Below it, read-only info badges show the registered email and member-since date. An editable form allows updating the name and mobile number. The 'Delete Account' button triggers a Bootstrap modal with a warning before permanent deletion.

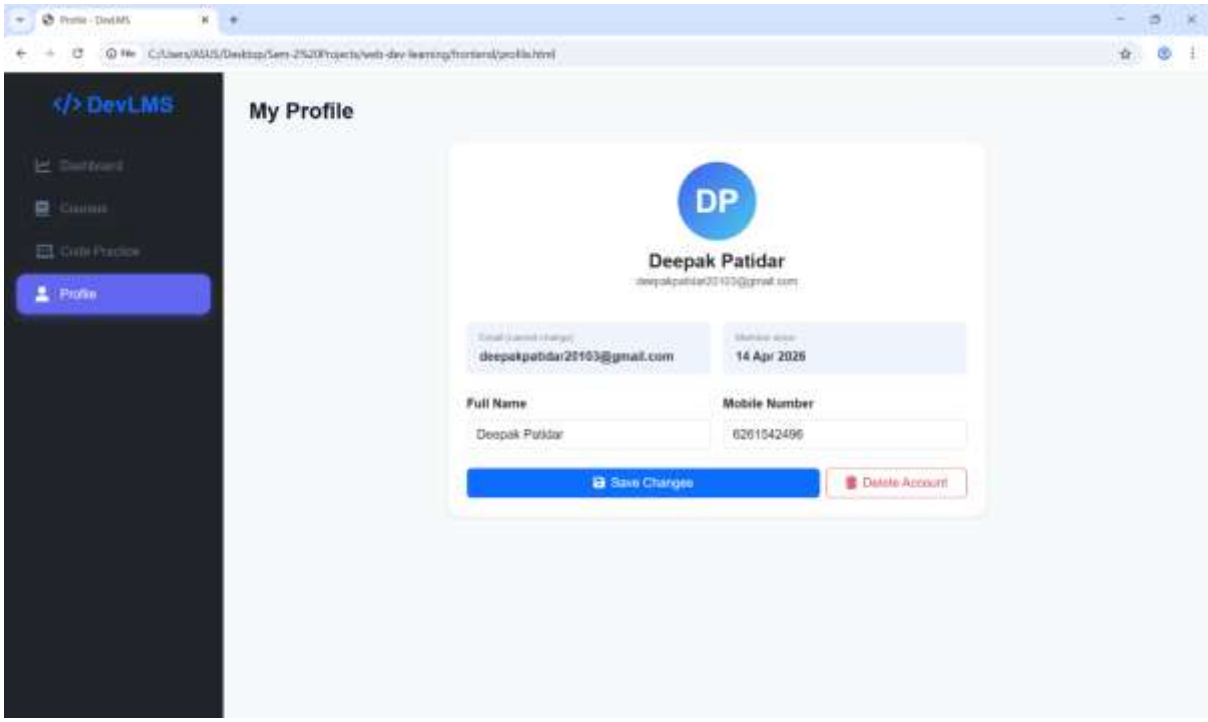

Figure 7 Profile Page

6.4 Courses Grid (courses.html)

The courses page renders all available courses as interactive cards in a responsive 3-column Bootstrap grid. Each card displays a thumbnail icon, course title, description excerpt, category badge, and a 'Start Learning' button. A real-time search input filters courses client-side by matching against both title and description strings without any additional API calls.

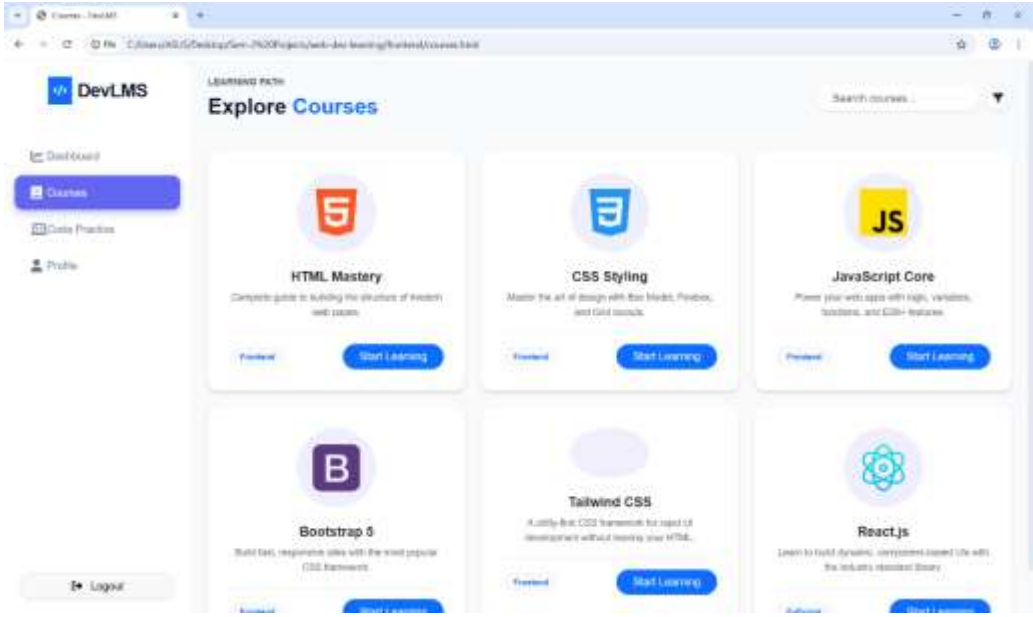

Figure 8 Courses Grid

7. LEARNING OUTCOMES

The development of the DevLMS project provided extensive hands-on learning across multiple dimensions of frontend web development. The following competencies were acquired and strengthened throughout the project lifecycle:

7.1 Core Frontend Skills

- Mastered semantic HTML5 document structuring with accessibility-aware markup.
- Advanced CSS3 techniques: custom properties (variables), glassmorphism, keyframe animations, and media queries.
- Implemented dark mode theming using CSS custom property overrides on a data-theme attribute.
- Built responsive layouts using Bootstrap 5 grid system (col-md-4, row g-4) and flexbox utilities.
- Applied CSS transitions and transform effects for interactive hover states on course cards.

7.2 JavaScript & API Integration

- Used async/await with the Fetch API to integrate RESTful backend endpoints across all pages.
- Built a centralized api.js module for DRY (Don't Repeat Yourself) API communication with token injection.
- Managed JWT authentication tokens using localStorage and implemented route protection on page load.
- Implemented client-side search filtering without DOM manipulation anti-patterns using innerHTML rerendering.
- Used URLSearchParams to read query string parameters for dynamic page routing (course ID, topic ID).
- Applied event delegation and DOM traversal for quiz option selection highlighting.

7.3 UI/UX Design Principles

- Learned the glassmorphism design pattern using backdrop-filter, semi-transparent backgrounds, and subtle borders.
- Designed a consistent sidebar navigation with active state styling and smooth transition effects.
- Implemented progressive loading feedback using Bootstrap spinners and animated skeleton states.
- Used Bootstrap modals for destructive action confirmation (account deletion) instead of browser confirm().
- Applied the principle of visual hierarchy using font weight, size, and color contrast throughout the UI.

7.4 Software Engineering Practices

- Followed modular architecture by separating JS concerns into api.js, auth.js, and dashboard.js files.
- Practiced XSS prevention using a custom escapeHTML() utility function in the quiz module.
- Implemented graceful error handling with try/catch blocks and user-friendly error messages.
- Applied consistent naming conventions and code formatting across all HTML, CSS, and JS files.
- Used Git version control with meaningful commit messages throughout the development process.
- Tested cross-browser compatibility across Chrome, Firefox, and Edge browsers.

8. REFERENCES

The following resources were consulted and referenced during the design and development of the DevLMS project:

Official Documentation

- MDN Web Docs — HTML, CSS, JavaScript reference: https://developer.mozilla.org
- Bootstrap 5 Official Documentation: https://getbootstrap.com/docs/5.3/
- Font Awesome Icons Documentation: https://fontawesome.com/docs
- Google Fonts API Documentation: https://fonts.google.com
- Fetch API Specification (MDN): https://developer.mozilla.org/en-US/docs/Web/API/Fetch_API

Learning Resources

- The Odin Project — Full Stack JavaScript Curriculum: https://www.theodinproject.com
- freeCodeCamp — Responsive Web Design & JavaScript Algorithms: https://www.freecodecamp.org
- CSS Tricks — A Complete Guide to Flexbox & Grid: https://css-tricks.com
- JavaScript.info — The Modern JavaScript Tutorial: https://javascript.info
- W3Schools — HTML, CSS, Bootstrap Tutorials: https://www.w3schools.com

Social Media Share Links

- **LinkedIn :-** https://www.linkedin.com/posts/deepakpatidar370_frontenddevelopment-webdevelopment-studentproject-ugcPost-7449842984973520896-yZiL?utm_source=share&utm_medium=member_desktop&rcm=ACoAADwxC4gBULpfVXV6mNQqptvuswvphGPeYQ4
- **Github :-** https://github.com/DeepakPatel370/DevLMS-Learning-Management-System-.git

— End of Report —

DevLMS Frontend Technologies Mini Project | 2026